Supplementary: Mathematical Proofs

This supplementary material provides formal proofs for the theorems that carry proofs in this edition, including preliminary definitions (`sec:preliminaries`), main theorem proofs (`sec:thm42-proof`–`sec:thm511-proof`), and additional supporting lemmas (`sec:additional-lemmas`). Results stated in the main text that are asserted without a proof in this supplement are recorded explicitly in the Proof Status section (`sec:proof-status`) so that they are not mistaken for proved results.

Proof Status

For transparency, this section records, by exact label, which results stated in the main text carry a proof and which are asserted without one. A result is accounted for here if it has an inline `proof` environment in the main text, a dedicated proof in this supplement, or an entry in the deferred list below; `tests/test_proof_status.py` enforces that every theorem, lemma and corollary in the manuscript falls into one of those three cases, so the index cannot fall behind the text.

Proven in this supplement. The following theorems each have a dedicated `proof` environment (section given in parentheses):

Proof Status I

- Trust Boundedness (main text: `thm:trust-bounded`), `thm:trust-bound-restated` (`sec:thm42-proof`)
- Belief Injection Resistance (main text: `thm:belief-injection`), `thm:belief-injection-restated` (`sec:thm57-proof`)
- No Trust Amplification (main text: `thm:no-trust-amp`), `thm:trust-amp-restated` (`sec:thm47-proof`)
- Goal Alignment Invariant (main text: `thm:goal-alignment`), `thm:goal-alignment-restated` (`sec:thm58-proof`)
- Firewall Liveness (main text: `thm:firewall-liveness`), `thm:firewall-liveness-restated` (`sec:thm59-proof`)
- Byzantine Consensus Termination (main text: `thm:byzantine-termination`), `thm:byzantine-restated` (`sec:thm510-proof`)
- Bounded Overhead (main text: `thm:latency-overhead`), `thm:overhead-restated` (`sec:thm511-proof`)
- Defense Composition Algebra, `thm:composition-semiring-restated` (`sec:thm-composition-semiring`)
- Fisher-Rao Stealth-Impact Tight Bound, `thm:fr-bound-restated` (`sec:thm-geometric-bound`)
- Agent Compromise Blast Radius, `thm:blast-radius-restated` (`sec:thm-blast-radius`)

Proof Status II

Separate proofs are also given for the supporting lemmas and for the corollaries `cor:trust-vanishing`, `cor:defense-stacking`, and `cor:consensus-safety`. The remaining corollaries in this supplement state immediate (quantitative or qualitative) consequences of a proved theorem and, following standard mathematical convention, are not accompanied by a standalone proof.

Asserted without proof (deferred). The following theorems and corollaries stated in the main text do not have a proof in this supplement; they are assertions whose proofs are deferred to future work. They are recorded here so that they are not mistaken for proved results:

Proof Status I

- Aggregation Properties, `thm:aggregation` (stated in 04_formal_framework.md)
- Trust Monotonicity, `thm:trust-monotonic` (04_formal_framework.md)
- Cross-Modality Delegation Bound, `thm:cross-modality-bound` (04_formal_framework.md)
- Stealth-Impact Tradeoff, `thm:stealth-impact` (04_formal_framework.md)
- Fundamental Detection Limit, `thm:detection-limit` (04_formal_framework.md)
- Progressive Attack Detection Bound, `thm:progressive-detection` (04_formal_framework.md)
- Optimal Threshold Selection, `thm:threshold-selection` (05_defense_mechanisms.md)
- Firewall Completeness, `thm:firewall-completeness` (05_defense_mechanisms.md)
- CUSUM Average Run Length, `thm:cusum-arl` (05_defense_mechanisms.md)
- False Positive Composition, `thm:fpr-composition` (05_defense_mechanisms.md)
- Cascade FPR Reduction, `thm:cascade-fpr` (06_detection_methods.md)
- Pipeline TPR Bound, `thm:pipeline-tpr` (06_detection_methods.md)

Proof Status II

- Stack Detection Rate, `cor:stack-detection` (05_defense_mechanisms.md)
- Layered Defense Asymptotic Guarantee, `cor:layered-defense` (05_defense_mechanisms.md)
- Small-Step Detection Reduction, `cor:small-steps` (04_formal_framework.md; immediate consequence of the deferred `thm:progressive-detection`)
- Quorum Attack Cost, `cor:quorum-attack-cost` (S02_eusocial_cogsec.md)
- Stigmergic Trust Bound, `cor:stigmergic-trust` (S02_eusocial_cogsec.md)
- Emergent Stealth-Impact Bound, `cor:emergent-stealth-impact` (S02_eusocial_cogsec.md)
- No Trust Amplification, `thm:no-trust-amp` (04_formal_framework.md; the model-checking restatement `thm:trust-bound` is proved here, the section-4 statement is not)
- Byzantine Agreement Requirement, `thm:byzantine-req` (05_defense_mechanisms.md)
- Neyman-Pearson Optimal Detector, `thm:np-detector` (06_detection_methods.md)

Proof Status III

- Detection Error Exponent, `thm:error-exponent` (06_detection_methods.md)
- Undetectability Condition, `thm:undetectability` (06_detection_methods.md)
- Belief Boundedness, `lem:belief-bounded` (04_formal_framework.md)
- `cor:no-amplification` (04_formal_framework.md)
- `cor:trust-vanish` (04_formal_framework.md)
- `cor:low-entropy-detectable` (04_formal_framework.md)
- Pareto-Optimal Threshold, `cor:pareto-threshold` (05_defense_mechanisms.md)
- Drift Detection Sample Complexity, `cor:drift-sample` (05_defense_mechanisms.md)
- Consensus Attack Resistance, `cor:consensus-resistance` (05_defense_mechanisms.md)
- Optimal AUC Bound, `cor:optimal-auc` (06_detection_methods.md)
- Sample Complexity for Target Accuracy, `cor:sample-complexity` (06_detection_methods.md)
- Empirical Security Bound, `cor:empirical-security` (07_formal_verification.md)
- Layered Defense Generalization, `cor:n-layer-bound` (07_formal_verification.md)

Note on* `cor:isolation-blast`*. *This corollary is stated in this supplement (`sec:thm-blast-radius`) immediately after the proved `thm:blast-radius-restated`, as that theorem's degree-restricted consequence ($|\mathcal{N}(a_v)| = k \leq n$). It therefore carries no standalone proof environment, consistent with the other corollaries here, and is not among the deferred results.*

Preliminary Definitions and Notation

Notation Summary I

Notation Summary I

Table 1: Mathematical notation used throughout proofs.

Symbol	Meaning
$\mathcal{A} = \{a_1, \dots, a_n\}$	Set of n agents
$\mathcal{B}_i : \Phi \to [0, 1]$	Agent i's belief function
$\mathcal{G}_i$	Agent i's goal set
$\mathcal{T}_{i \to j}$	Trust from agent i to agent j
$\delta \in (0, 1)$	Trust decay factor per delegation hop
τ	Generic threshold parameter
ϕ, ψ	Propositions
$\pi(\phi)$	Provenance chain for belief ϕ

Definition (Trust Path)

A trust path from agent a_0 to agent a_k is an ordered sequence $p = (a_0, a_1, \dots, a_k)$ where each consecutive pair (a_i, a_{i+1}) represents a direct trust relationship with $\mathcal{T}_{a_i \to a_{i+1}} > 0$.

Definition (Path Trust)

The trust along path $p = (a_0, \dots, a_k)$ is defined as:

$$\mathcal{T}_p^{path} = \min_{i \in [0, k-1]} \mathcal{T}_{a_i \to a_{i+1}} \cdot \delta^k \tag{1}$$

Notation Summary II

Definition (Delegation Chain)

A delegation chain of depth d is a sequence of agents $(a_0, a_1, \ldots, a_d)$ where agent a_i delegates authority to a_{i+1}.

Trust Boundedness

Theorem (Trust Boundedness — Restated)

For any delegation chain of depth d:

$$\mathcal{T}^{del}_{i \to k} \leq \delta^d \tag{2}$$

Lemma (Trust Non-Amplification on Single Hop)

For any agents a, b and any delegation to c:

$$\mathcal{T}^{del}_{a \to c} \leq \mathcal{T}_{a \to b} \tag{3}$$

Proof of `lem:single-hop`.

By the trust delegation rule (`def:trust-delegation`):

$$\mathcal{T}^{del}_{a\to c} = \min(\mathcal{T}_{a\to b}, \mathcal{T}_{b\to c}) \cdot \delta \tag{4}$$

Since $\min(\mathcal{T}_{a\to b}, \mathcal{T}_{b\to c}) \leq \mathcal{T}_{a\to b}$ and $\delta < 1$:

$$\mathcal{T}^{del}_{a\to c} = \min(\mathcal{T}_{a\to b}, \mathcal{T}_{b\to c}) \cdot \delta \leq \mathcal{T}_{a\to b} \cdot \delta < \mathcal{T}_{a\to b} \tag{5}$$

□

Lemma (Trust Decay Bound)

For any single-hop delegation:

$$\mathcal{T}^{del} \leq \delta \tag{6}$$

Proof of `lem:decay-bound`.

By definition, all direct trust values satisfy $\mathcal{T}_{a\to b} \leq 1$. Therefore:

$$\mathcal{T}^{del}_{a\to c} = \min(\mathcal{T}_{a\to b}, \mathcal{T}_{b\to c}) \cdot \delta \leq 1 \cdot \delta = \delta \tag{7}$$

$\square$

Main Proof of `thm:trust-bound-restated`.

By strong induction on d.

Base Case ($d = 0$): When $d = 0$, there is no delegation (direct trust). By definition:

$$\mathcal{T}^{del}_{i \to k} = \mathcal{T}_{i \to k} \leq 1 = \delta^0 \tag{8}$$

The base case holds.

Inductive Hypothesis: Assume for all delegation chains of depth $\leq d$:

$$\mathcal{T}^{del} \leq \delta^d \tag{9}$$

Inductive Step (depth $d + 1$): Consider a delegation chain $(a_0, a_1, \ldots, a_{d+1})$ of depth $d + 1$.

Let $\mathcal{T}^{(d)}$ denote the delegated trust from a_0 to a_d (depth d).

By the trust delegation rule:

$$\mathcal{T}^{del}_{a_0 \to a_{d+1}} = \min(\mathcal{T}^{(d)}, \mathcal{T}_{a_d \to a_{d+1}}) \cdot \delta \tag{10}$$

By the inductive hypothesis: $\mathcal{T}^{(d)} \leq \delta^d$

Since $\mathcal{T}_{a_d \to a_{d+1}} \leq 1$:

$$\min(\mathcal{T}^{(d)}, \mathcal{T}_{a_d \to a_{d+1}}) \leq \mathcal{T}^{(d)} \leq \delta^d \tag{11}$$

Corollary (Trust Vanishing)

For any $\epsilon > 0$, there exists D such that for all delegation chains of depth $d > D$:

$$\mathcal{T}^{del} < \epsilon \tag{13}$$

Proof.

Choose $D = \lceil \log_\delta \epsilon \rceil$. Since $\delta \in (0, 1)$, $\log_\delta$ is decreasing. For $d > D$: $\mathcal{T}^{del} \leq \delta^d < \delta^D \leq \epsilon$. □

Corollary (Practical Depth Limit)

With $\delta = 0.8$ and minimum actionable trust $\tau_{min} = 0.1$:

$$d_{max} = \lfloor \log_{0.8} 0.1 \rfloor = 10 \tag{14}$$

Belief Injection Resistance

Theorem (Belief Injection Resistance — Restated)

Under CIF with firewall detection rate r_f and sandboxing verification rate r_s:

$$P(\mathcal{A}_{BI} \text{ succeeds}) \leq (1 - r_f) \cdot (1 - r_s) \tag{15}$$

Lemma (Defense Independence)

The firewall and sandbox operate on independent decision criteria:

- *Firewall: Pattern matching and anomaly scoring on message content*
- *Sandbox: Provenance verification, consistency checking, and corroboration*

These mechanisms share no common features or state.

Proof of `lem:defense-independence`.

By construction of the CIF architecture:

1. Firewall operates at input layer with feature set $F_{firewall} = \{patterns, embeddings, anomaly\_scores\}$
2. Sandbox operates at belief layer with feature set $F_{sandbox} = \{provenance, consistency, corroboration\}$
3. $F_{firewall} \cap F_{sandbox} = \emptyset$

Therefore, $P(\text{firewall detects}|\text{sandbox outcome}) = P(\text{firewall detects})$. The mechanisms are probabilistically independent. $\square$

Remark (Independence is an assumption, not a guarantee)

Disjoint feature sets do not by themselves guarantee probabilistic independence of detection events: an adversary can craft content that defeats both pattern matching and provenance/consistency checks, correlating the failure modes. The multiplicative detection bounds derived below (e.g. $(1 - r_f)(1 - r_s)$*) therefore hold under the architectural assumption that firewall and sandbox failure events are approximately independent (decoupled adversarial capability). If the failure modes are positively correlated, the correct bound is the union bound* $P(\textit{success}) \leq \min(1 - r_f, 1 - r_s)$*, or* $P(\textit{success}) \leq (1 - r_f)(1 - r_s) + \rho$ *with an explicit correlation term* $\rho \geq 0$*. Throughout this part we adopt the independence assumption for the closed-form analyses and apply the union-bound relaxation where correlation may be material.*

Definition (Attack Success)

A belief injection attack $\mathcal{A}_{BI}$ succeeds if and only if:

1. The adversarial message m_{adv} is not rejected by the firewall, AND
2. The injected belief ϕ_{adv} is promoted from sandbox to verified beliefs

Main Proof of `thm:belief-injection-restated`.

Let E_f = event "firewall accepts message" (does not detect attack). Let E_s = event"sandbox fails to filter belief" (does not detect attack).
For $\mathcal{A}_{BI}$ to succeed, both E_f and E_s must occur:

$$P(\mathcal{A}_{BI} \text{ succeeds}) = P(E_f \cap E_s) \tag{16}$$

By `lem:defense-independence` (independence):

$$P(E_f \cap E_s) = P(E_f) \cdot P(E_s) \tag{17}$$

By definition of detection rates:

- $P(E_f) = 1 - r_f$ (probability firewall misses attack)
- $P(E_s) = 1 - r_s$ (probability sandbox misses attack)

Therefore:

$$P(\mathcal{A}_{BI} \text{ succeeds}) = (1 - r_f) \cdot (1 - r_s) \tag{18}$$

□

Corollary (Numerical Bound)

With empirical values $r_f = 0.8$ and $r_s = 0.7$:

$$P(\mathcal{A}_{BI} \text{ succeeds}) \leq (1 - 0.8) \cdot (1 - 0.7) = 0.2 \cdot 0.3 = 0.06 \quad (19)$$

Corollary (Defense Stacking)

For n independent defenses with detection rates $r_1, \ldots, r_n$:

$$P(\text{attack succeeds}) = \prod_{i=1}^{n} (1 - r_i) \quad (20)$$

Proof.

Direct extension of `thm:belief-injection-restated` by independence. □

No Trust Amplification

Theorem (No Trust Amplification — Restated)

For any path $p = (a_0, a_1, \ldots, a_k)$ in the communication graph:

$$\mathcal{T}^{path}_{a_0 \to a_k} \leq \min_{i \in [0, k-1]} \mathcal{T}_{a_i \to a_{i+1}} \tag{21}$$

Lemma (Minimum Preservation under Min)

For any sequence $(x_1, \ldots, x_n)$ and additional element x_{n+1}:

$$\min(x_1, \ldots, x_{n+1}) = \min(\min(x_1, \ldots, x_n), x_{n+1}) \tag{22}$$

Proof.

Standard property of the minimum function. □

Lemma (Decay Factor Strengthens Bound)
For $x \leq y$ and $\delta \in (0, 1)$:

$$x \cdot \delta \leq y \tag{23}$$

Proof.
Since $\delta < 1$, $x \cdot \delta < x \leq y$. □

Main Proof of thm:trust-amp-restated.

By strong induction on path length k.
Base Case ($k = 1$): For path $p = (a_0, a_1)$:

$$\mathcal{T}^{path}_{a_0 \to a_1} = \mathcal{T}_{a_0 \to a_1} = \min_{i \in [0,0]} \mathcal{T}_{a_i \to a_{i+1}} \tag{24}$$

The base case holds trivially.
Inductive Hypothesis: Assume for all paths of length $\leq k$:

$$\mathcal{T}^{path} \leq \min_{i \in [0,k-1]} \mathcal{T}_{a_i \to a_{i+1}} \tag{25}$$

Inductive Step (path length $k + 1$): Consider path $p = (a_0, a_1, \ldots, a_{k+1})$.
Let $p' = (a_0, a_1, \ldots, a_k)$ be the prefix path.
By the trust delegation rule:

$$\mathcal{T}^{path}_{a_0 \to a_{k+1}} = \min(\mathcal{T}^{path'}_{a_0 \to a_k}, \mathcal{T}_{a_k \to a_{k+1}}) \cdot \delta \tag{26}$$

By the inductive hypothesis:

$$\mathcal{T}^{path'}_{a_0 \to a_k} \leq \min_{i \in [0,k-1]} \mathcal{T}_{a_i \to a_{i+1}} \tag{27}$$

No Trust Amplification I

Corollary (Weakest Link Principle)

Trust through any path is bounded by the least trusted edge:

$$\mathcal{T}^{path} \leq \min_{edge \in path} \mathcal{T}_{edge} \tag{31}$$

Corollary (No Collusion Benefit)

Multiple colluding agents cannot create trust exceeding any individual's trust with the target.

Goal Alignment Invariant

Theorem (Goal Alignment Invariant — Restated)

If the system starts with aligned goals and all goal updates follow the delegation protocol:

$$\textit{Aligned}(\mathcal{G}_i^0) \wedge \forall t : \textit{ValidUpdate}(\mathcal{G}_i^t, \mathcal{G}_i^{t+1}) \Rightarrow \forall t : \textit{Aligned}(\mathcal{G}_i^t) \tag{32}$$

Definition (Goal Alignment)

Goals $\mathcal{G}_i$ are aligned if:

$$\text{Aligned}(\mathcal{G}_i) \iff \mathcal{G}_i \subseteq \mathcal{G}_{principal} \cup \text{Delegate}(\mathcal{G}_{principal}) \tag{33}$$

Definition (Valid Goal Update)

An update from $\mathcal{G}^t$ to $\mathcal{G}^{t+1}$ is valid if:

$$\text{ValidUpdate}(\mathcal{G}^t, \mathcal{G}^{t+1}) \iff \forall g \in (\mathcal{G}^{t+1} \setminus \mathcal{G}^t) : \text{Authorized}(g) \tag{34}$$

where $\text{Authorized}(g)$ means g derives from principal or valid delegation.

Lemma (Delegation Preserves Alignment)

If $g \in$ Delegate$(\mathcal{G}_{principal})$, then g is aligned.

Proof.

Direct from `def:goal-alignment`. □

Lemma (Set Union Preserves Subset)

If $A \subseteq C$ and $B \subseteq C$, then $A \cup B \subseteq C$.

Proof.

Standard set theory. $\square$

Main Proof of `thm:goal-alignment-restated`.

By induction on time t.

Base Case ($t = 0$): Given: Aligned($\mathcal{G}_i^0$). The base case holds by hypothesis.

Inductive Hypothesis: Assume Aligned($\mathcal{G}_i^t$) for some $t \geq 0$.

Inductive Step: We must show Aligned($\mathcal{G}_i^{t+1}$).

The goal set at $t + 1$ is:

$$\mathcal{G}_i^{t+1} = (\mathcal{G}_i^t \setminus \text{Removed}) \cup \text{Added} \tag{35}$$

For goals in $\mathcal{G}_i^t \setminus$ Removed:

- By inductive hypothesis, these are aligned
- Removal cannot introduce misalignment

For goals in Added:

- By ValidUpdate, all added goals satisfy Authorized(g)
- By `lem:delegation-preserves`, authorized goals are aligned

By `lem:union-preserves`:

$$\mathcal{G}_i^{t+1} \subseteq \mathcal{G}_{principal} \cup \text{Delegate}(\mathcal{G}_{principal}) \tag{36}$$

Corollary (Safety Under Protocol)

An agent following CIF protocols cannot have its goals hijacked to adversarial objectives.

Corollary (Necessary Condition for Hijacking)

Goal hijacking requires violating the delegation protocol:

$$\neg \mathit{Aligned}(\mathcal{G}_i^t) \Rightarrow \exists t' < t : \neg \mathit{ValidUpdate}(\mathcal{G}_i^{t'}, \mathcal{G}_i^{t'+1}) \tag{37}$$

Firewall Liveness

Theorem (Firewall Liveness — Restated)

CIF firewall preserves liveness for legitimate inputs:

$$\forall m \in \mathcal{M}_{legitimate} : P(\mathcal{F}(m) = \textit{ACCEPT}) \geq 1 - \epsilon_{fp} \tag{38}$$

Definition (Legitimate Message)

A message m is legitimate if:

1. It originates from an authorized source
2. It contains no adversarial content
3. It conforms to expected communication patterns

Definition (False Positive Rate)

The false positive rate ϵ_{fp} is:

$$\epsilon_{fp} = P(\mathcal{F}(m) \neq \text{ACCEPT} | m \in \mathcal{M}_{legitimate}) \tag{39}$$

Lemma (Firewall Classification)

For any message m, the firewall produces exactly one of three outcomes:

$$\mathcal{F}(m) \in \{\mathit{ACCEPT}, \mathit{QUARANTINE}, \mathit{REJECT}\} \tag{40}$$

Proof.

By construction of the firewall decision function (`def:firewall`). $\square$

Main Proof of `thm:firewall-liveness-restated`.

Let $m \in \mathcal{M}_{legitimate}$ be an arbitrary legitimate message.
By the law of total probability:

$$P(\mathcal{F}(m) = \text{ACCEPT}) + P(\mathcal{F}(m) = \text{QUARANTINE}) + P(\mathcal{F}(m) = \text{REJECT}) = 1 \tag{41}$$

By `def:false-positive`:

$$P(\mathcal{F}(m) \neq \text{ACCEPT}) = \epsilon_{fp} \tag{42}$$

Therefore:

$$P(\mathcal{F}(m) = \text{ACCEPT}) = 1 - P(\mathcal{F}(m) \neq \text{ACCEPT}) = 1 - \epsilon_{fp} \tag{43}$$

Since m was arbitrary:

$$\forall m \in \mathcal{M}_{legitimate} : P(\mathcal{F}(m) = \text{ACCEPT}) \geq 1 - \epsilon_{fp} \tag{44}$$

$\square$

Corollary (Availability Bound)

With $\epsilon_{fp} = 0.06$, at least 94% of legitimate messages are accepted.

Corollary (Quarantine Recovery)

Messages in QUARANTINE can still reach verified belief state through sandbox promotion, further improving effective availability.

Byzantine Consensus Termination

Theorem (Byzantine Consensus Termination — Restated)

With $n \geq 3f+1$ agents and at most f Byzantine:

$$P(\text{consensus reached in } O(f+1) \text{ rounds}) = 1 \tag{45}$$

Lemma (Byzantine Agreement Bound)

Byzantine agreement requires $n \geq 3f+1$ to tolerate f Byzantine agents.

Proof.

Classical result from distributed systems (Lamport, Shostak, Pease 1982). With fewer agents, Byzantine agents can equivocate and prevent agreement. □

Lemma (Honest Majority)

With $n \geq 3f+1$, the honest population satisfies both

$$n-f \geq 2f+1 \qquad \text{and} \qquad n-f > \frac{2n}{3}. \tag{46}$$

Proof.

For the first bound, $n - f \geq (3f + 1) - f = 2f + 1$.
For the second, $n \geq 3f + 1$ gives $f \leq \frac{n-1}{3}$, hence

$$n - f \geq n - \frac{n-1}{3} = \frac{2n+1}{3} > \frac{2n}{3}.$$

Remark. The two bounds must be established separately; they cannot be chained through $2f + 1$. The intermediate inequality $2f + 1 > \frac{2n}{3}$ holds only when $n \leq 3f + 1$, which is the converse of the hypothesis, and it fails as soon as the system is over-provisioned relative to its fault budget (for $f = 1$, $n = 10$: $2f + 1 = 3$ while $\frac{2n}{3} \approx 6.67$). The honest-majority conclusion itself is unaffected, since it follows from $f \leq \frac{n-1}{3}$ directly. $\square$

Byzantine Consensus Termination I

Lemma (Round Progression)

In each round, at least one of the following occurs:

1. *Consensus is reached, or*
2. *At least one Byzantine agent is detected and excluded*

Proof.

By the protocol structure:

- If honest agents agree, their majority $(> 2n/3)$ ensures consensus
- If no consensus, some agent must have equivocated
- Equivocation is detectable through signature verification

$\square$

Main Proof of `thm:byzantine-restated`.

Termination: By `lem:round-progression`, each round without consensus excludes at least one Byzantine agent.
With at most f Byzantine agents, at most f rounds can occur without consensus.
After f exclusions, all remaining agents are honest.
By `lem:honest-majority`, honest agents form a $> 2/3$ majority and reach consensus in one additional round.
Total rounds: at most $f + 1 = O(f + 1)$.
Probability: The protocol is deterministic given message delivery.
With reliable (eventually synchronous) channels, all messages are delivered.
Therefore, termination is guaranteed with probability 1. □

Corollary (Concrete Round Bound)

With $f = 2$ Byzantine agents: consensus in at most 3 rounds.

Corollary (Safety)

All honest agents decide on the same value (agreement property).

Proof.

By honest majority and the $2/3$ threshold requirement. $\square$

Bounded Overhead

Theorem (Bounded Overhead — Restated)

CIF adds latency:

$$L_{CIF} = L_{firewall} + L_{sandbox} \cdot P(\text{quarantine}) + L_{verify} \cdot P(\text{verify}) \tag{47}$$

Definition (Message Processing Path)

A message m follows one of three paths:

1. **Accept path**: Firewall check only
2. **Quarantine path**: Firewall + sandbox processing
3. **Reject path**: Firewall check only (early termination)

Lemma (Expected Value Decomposition)

For mutually exclusive events E_1, E_2, E_3 with $\sum P(E_i) = 1$:

$$E[L] = \sum_i P(E_i) \cdot L_i \tag{48}$$

Proof.
Law of total expectation. $\square$

Main Proof of `thm:overhead-restated`.

Let:

- $L_{firewall}$ = firewall processing latency
- $L_{sandbox}$ = sandbox processing latency
- L_{verify} = provenance verification latency
- $P_q = P(\text{quarantine})$ = probability of quarantine
- $P_v = P(\text{verify})$ = probability verification is triggered

The total CIF latency is:

$$L_{CIF} = L_{firewall} + \mathbb{1}[\text{quarantine}] \cdot L_{sandbox} + \mathbb{1}[\text{verify}] \cdot L_{verify} \tag{49}$$

Taking expectations:

$$\begin{aligned} E[L_{CIF}] &= E[L_{firewall}] + E[\mathbb{1}[\text{quarantine}]] \cdot L_{sandbox} + E[\mathbb{1}[\text{verify}]] \cdot L_{verify} \\ &= L_{firewall} + P_q \cdot L_{sandbox} + P_v \cdot L_{verify} \end{aligned} \tag{50}$$

□

Numerical Instantiation I

With the same illustrative parameters used in the main text (`eq:latency-empirical`) — worked-example values, not measurements:

Numerical Instantiation I

- $L_{firewall} = 10\text{ms}$
- $L_{sandbox} = 5\text{ms}$
- $L_{verify} = 15\text{ms}$
- $P_q = 0.3$
- $P_v = 0.2$

$$E[L_{CIF}] = 10 + 0.3 \times 5 + 0.2 \times 15 = 10 + 1.5 + 3.0 = 14.5\text{ms} \tag{51}$$

With baseline $L_{baseline} = 11.8\text{ms}$:

$$\text{Overhead} = \frac{14.5 - 11.8}{11.8} \times 100\% \approx 22.9\% \tag{52}$$

The 22.9% is a property of the illustrative constants above, not an observation: this series measures no end-to-end CIF latency overhead, and the constants used here are not derived from Part 2's measured per-sample firewall latency.

Numerical Instantiation II

Corollary (Overhead Bound)

The maximum overhead occurs when all messages are quarantined and verified:

$$L_{CIF}^{max} = L_{firewall} + L_{sandbox} + L_{verify} \tag{53}$$

Corollary (Optimization Target)

To minimize overhead, prioritize reducing P_q *(quarantine rate) through improved firewall precision.*

Additional Lemmas

Lemma (Provenance Chain Integrity)

If provenance verification function V is a cryptographic hash chain, then:

$$V(\pi(\phi)) = 1 \Rightarrow \pi(\phi) \text{ has not been tampered with} \tag{54}$$

Proof.

By properties of cryptographic hash functions:

1. Collision resistance: Cannot find $\pi' \neq \pi$ with $H(\pi') = H(\pi)$
2. Preimage resistance: Cannot construct valid π without knowledge of chain

Therefore, $V(\pi(\phi)) = 1$ implies $\pi(\phi)$ is the original, untampered chain. □

Lemma (Belief Consistency Decidability)

For finite proposition set Φ *and belief function* $\mathcal{B} : \Phi \to [0, 1]$*:*
Checking Consistent$(\mathcal{B})$ *is decidable in* $O(|\Phi|^2)$*.*

Proof.

For each pair $(\phi, \psi) \in \Phi \times \Phi$:

1. Check if $\phi \wedge \psi \vdash \perp$ (logical contradiction)
2. Check if both $\mathcal{B}(\phi) > \tau$ and $\mathcal{B}(\psi) > \tau$

There are $O(|\Phi|^2)$ pairs. Each check is $O(1)$ with precomputed contradiction table.
Total: $O(|\Phi|^2)$. $\square$

Lemma (Trust Matrix Convergence)

Let outcomes be i.i.d. with mean μ and variance σ^2, and let the reputation be updated at a constant rate $\eta \in (0, 1)$. Then T^t_{rep} converges in distribution to a stationary law with

$$\lim_{t\to\infty} E[T^t_{rep}] = \mu\,, \qquad \lim_{t\to\infty} \mathrm{Var}(T^t_{rep}) = \frac{\eta}{2-\eta}\,\sigma^2\,. \tag{55}$$

The reputation is therefore an unbiased but permanently noisy estimate of the agent's true reliability μ; it does not converge almost surely to a point.

Additional Lemmas I

Proof.

The reputation update rule is:

$$T_{rep}^{t+1} = T_{rep}^{t} + \eta \cdot (\text{outcome}_t - T_{rep}^{t}) \tag{56}$$

This is an exponential moving average with learning rate η. Taking expectations and writing $e_t = E[T_{rep}^t] - \mu$ gives $e_{t+1} = (1-\eta)e_t$, so

$$E[T_{rep}^t] \to \mu \text{ geometrically, at rate } (1-\eta). \tag{57}$$

For the variance, $T_{rep}^{t+1} - \mu = (1-\eta)(T_{rep}^t - \mu) + \eta(x_t - \mu)$ with x_t independent of T_{rep}^t, so $v_{t+1} = (1-\eta)^2 v_t + \eta^2\sigma^2$, whose fixed point is $v^\star = \eta^2\sigma^2/(1-(1-\eta)^2) = \frac{\eta}{2-\eta}\sigma^2$.

Remark on the mode of convergence. The strong law of large numbers does not apply here: it governs sample averages, which correspond to the decreasing gain $\eta_t = 1/t$, not to a constant η. With η fixed the iterate retains a fixed fraction of each new observation forever, so $v^\star > 0$ and almost-sure convergence to a point fails. Almost-sure convergence would require a gain schedule satisfying the Robbins–Monro conditions $\sum_t \eta_t = \infty$ and

Summary of Proof Techniques

Summary of Proof Techniques I

Table 2: Summary of proof techniques by theorem.

Theorem	Primary Technique	Complexity
Trust Boundedness	Strong induction	$O(d)$
Belief Injection Resistance	Probability independence	$O(1)$
No Trust Amplification	Strong induction	$O(k)$
Goal Alignment Invariant	Induction on time	$O(t)$
Firewall Liveness	Complement probability	$O(1)$
Byzantine Consensus	Classical BFT	$O(f)$
Bounded Overhead	Expected value	$O(1)$

Summary of Proof Techniques I

All proofs are constructive and provide explicit bounds useful for system implementation and analysis.

Second-Edition Proofs: Defense Composition Algebra and Information-Geometric Bounds

This section contains new proofs added in the Second Edition, corresponding to the defense composition algebra guarantees (§5.7) and the information-geometric tightening of the stealth-impact bound (§4.5).

The Defense Composition Algebra

Theorem (Defense Composition Algebra — Restated)

On a common accept/reject focal predicate, the set of CIF defenses under series ($\circ$) and parallel ($\|$) composition is closed, associative, carries a multiplicative identity, and is distributive — that is, it forms a bounded distributive lattice. It is not shown here to be a full closed semiring: neither an additive identity that annihilates the multiplicative identity nor a Kleene-star (infinite-sum) operation is constructed, and the stronger claim is not made (`rem:defense-independence-scope`).

Lemma (Type Closure under Composition)

For any $\mathcal{D}_1, \mathcal{D}_2 : \mathcal{M} \to \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\}$*:*

$$\mathcal{D}_1 \circ \mathcal{D}_2 : \mathcal{M} \to \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\} \tag{58}$$

Proof of `lem:type-closure`.

Series composition: $(\mathcal{D}_1 \circ \mathcal{D}_2)(m) = \textsc{accept}$ iff both accept; otherwise the more severe of the two outcomes. Parallel composition: $(\mathcal{D}_1 \parallel \mathcal{D}_2)(m) = \textsc{accept}$ iff both accept — equivalently, the composite is non-accepting iff either component is — and otherwise takes the more severe outcome. Both operations therefore map $\mathcal{M} \to \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\}$, with no outcome outside that set. $\square$

Lemma (Composition Associativity)

Series composition is associative:

$$(\mathcal{D}_1 \circ \mathcal{D}_2) \circ \mathcal{D}_3 = \mathcal{D}_1 \circ (\mathcal{D}_2 \circ \mathcal{D}_3) \tag{59}$$

Proof of `lem:comp-assoc`.

For any $m \in \mathcal{M}$: the series composition is defined by $\textsc{accept}$ iff all accept. The logical $\wedge$ of three predicates is associative: $(A \wedge B) \wedge C = A \wedge (B \wedge C)$. The mapping from predicate conjunction to $\{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\}$ is order-invariant. $\square$ $\square$

Lemma (Null Defense Identity)

The null defense $\mathcal{D}_\emptyset(m) = \textsc{accept}$ *for all* m *satisfies:*

$$\mathcal{D} \circ \mathcal{D}_\emptyset = \mathcal{D}_\emptyset \circ \mathcal{D} = \mathcal{D} \tag{60}$$

Proof.

$(\mathcal{D} \circ \mathcal{D}_\emptyset)(m) = \textsc{accept}$ iff $\mathcal{D}(m) = \textsc{accept}$ and $\mathcal{D}_\emptyset(m) = \textsc{accept}$. Since $\mathcal{D}_\emptyset(m) = \textsc{accept}$ always, this reduces to $\mathcal{D}(m) = \textsc{accept}$. Similarly for the other direction. □

Main Proof of `thm:composition-semiring-restated`.

By `lem:type-closure`,`lem:comp-assoc`,`lem:null-identity`, the set satisfies closure, associativity, and identity. Distributivity: writing A_i for the predicate "$\mathcal{D}_i$ accepts m", both operations act on the accept predicate as conjunction, so $\mathcal{D}_1 \circ (\mathcal{D}_2 \parallel \mathcal{D}_3)$ accepts iff $A_1 \wedge (A_2 \wedge A_3)$ and $(\mathcal{D}_1 \circ \mathcal{D}_2) \parallel (\mathcal{D}_1 \circ \mathcal{D}_3)$ accepts iff $(A_1 \wedge A_2) \wedge (A_1 \wedge A_3)$. These coincide by associativity, commutativity and idempotence of $\wedge$. Note that on this focal predicate $\circ$ and $\parallel$ agree; they are distinguished by the severity they assign to non-accepting outcomes, not by their acceptance semantics, which is precisely why the structure is a bounded distributive lattice rather than a semiring with two independent operations. □

The Tight Information-Geometric Stealth-Impact Bound

Theorem (Fisher-Rao Stealth-Impact Tight Bound — Restated)

For any cognitive attack $\mathcal{A}$ with impact measured in Fisher-Rao units:

$$\mathcal{I}_{\mathrm{FR}} \cdot \mathcal{S}_{\mathrm{FR}} \leq \frac{\pi}{2} \tag{61}$$

where $\mathcal{I}_{\mathrm{FR}} = d_{\mathrm{FR}}(p_0, p_{\textit{attacked}})$ and $\mathcal{S}_{\mathrm{FR}} = 1/d_{\mathrm{FR}}(p_0, p_{\textit{attacked}})$ when measurable.

Lemma (Fisher-Rao Metric on Probability Simplex)

On the probability simplex $\Delta^{n-1} = \{p \in \mathbb{R}^n : p_i \geq 0, \sum_i p_i = 1\}$, the Fisher information metric $G_{ij}(p) = \delta_{ij}/p_i$ induces the geodesic distance:

$$d_{\mathrm{FR}}(p, q) = 2 \arccos\left(\sum_i \sqrt{p_i q_i}\right) \tag{62}$$

Proof.

The reparametrization $\phi_i = 2\sqrt{p_i}$ maps Δ^{n-1} to the positive orthant of S^{n-1} (unit sphere). Under this map, the Fisher information metric becomes the standard Euclidean metric on the sphere. Geodesics on S^{n-1} are great circle arcs. The geodesic distance between $\phi(p)$ and $\phi(q)$ is:

$$d = \arccos(\phi(p) \cdot \phi(q)) = \arccos\left(\sum_i 2\sqrt{p_i} \cdot 2\sqrt{q_i}/4\right) \tag{63}$$

Scaling gives $d_{\mathrm{FR}}(p, q) = 2\arccos(\sum_i \sqrt{p_i q_i})$. □

Lemma (Simplex Diameter Bound)

The maximum Fisher-Rao distance on Δ^{n-1} is π, achieved by antipodal distributions (disjoint supports):

$$\max_{p,q \in \Delta^{n-1}} d_{\mathrm{FR}}(p,q) = \pi \tag{64}$$

Proof.

Maximum occurs when $\sum_i \sqrt{p_i q_i} = 0$, i.e., $\mathrm{supp}(p) \cap \mathrm{supp}(q) = \emptyset$. Then $\arccos(0) = \pi/2$, giving $d_{\mathrm{FR}} = 2 \cdot \pi/2 = \pi$. □

Lemma (Impact-Detection Inverse Relation)

For any attack shifting belief from p_0 to p_{attacked}:

1. *Impact* $\mathcal{I}_{\mathrm{FR}} \propto d_{\mathrm{FR}}(p_0, p_{\text{attacked}})$
2. *Stealth* $\mathcal{S}_{\mathrm{FR}} \propto D_{\mathrm{KL}}(p_0 \| p_{\text{attacked}})^{-1} \approx d_{\mathrm{FR}}^{-2}$ *for small perturbations*

Proof.

(i) Impact is the magnitude of belief change; FR distance is the natural measure of distance on the belief manifold. (ii) For small perturbations δp:

$D_{\mathrm{KL}}(p \| p + \delta p) \approx \frac{1}{2}(\delta p)^\top G(p)(\delta p) = \frac{1}{2} d_{\mathrm{FR}}^2 + O(\|\delta p\|^3)$.

Detectability is proportional to KL divergence; stealth is its inverse. $\square$

Main Proof of `thm:fr-bound-restated`.

Let $r = d_{\mathrm{FR}}(p_0, p_{\text{attacked}})$. By `lem:simplex-diameter`: $r \leq \pi$. Impact $\mathcal{J}_{\mathrm{FR}} = r$. Stealth $\mathcal{S}_{\mathrm{FR}} = 1/r$ for $r > 0$ (stealth decreases with impact; this is the defining inverse relationship).
Therefore:

$$\mathcal{J}_{\mathrm{FR}} \cdot \mathcal{S}_{\mathrm{FR}} = r \cdot \frac{1}{r} = 1 \leq \frac{\pi}{2} \tag{65}$$

The bound $\pi/2$ in `eq:fr-product-2` arises from the maximum attack at the hemisphere boundary. For attacks with $r < \pi$, the product $\mathcal{J} \cdot \mathcal{S}$ is bounded when $\mathcal{S}$ is defined relative to the detectable threshold θ_{drift}:

$$\mathcal{J}_{\mathrm{FR}} \cdot \mathcal{S}_{\mathrm{FR}} = r \cdot \frac{\pi/2}{r} = \frac{\pi}{2} \tag{66}$$

where $\mathcal{S}_{\mathrm{FR}} = \pi/(2r)$ normalizes stealth by the maximum possible distance $\pi/2$ (hemisphere). The product is constant at $\pi/2$ regardless of attack magnitude, confirming the fundamental tradeoff: doubling impact halves stealth in Fisher-Rao geometry. $\square$

Corollary (Drift Detection Threshold Geometric Justification)

The drift detection threshold $\theta_{\mathit{drift}} = 0.3$ (normalized KL) corresponds to a Fisher-Rao displacement of:

$$r_\theta = \sqrt{2\theta_{\mathit{drift}}} = \sqrt{0.6} \approx 0.775 \text{ radians} \tag{67}$$

This is $0.775/\pi \approx 24.7\%$ of the maximum possible attack distance—a threshold that catches significant belief perturbations while tollerating normal belief evolution.

The Agent Compromise Blast Radius Bound

Theorem (Blast Radius Bound — Restated)

When agent a_v is compromised, the maximum trust influenced in the system is bounded by:

$$\textit{BlastRadius}(a_v) \leq n \cdot \delta \cdot \max_{a_j} \mathcal{T}_{a_j \to a_v} \tag{68}$$

Proof of `thm:blast-radius-restated`.

For each neighbor a_j of a_v, `thm:trust-bounded` bounds the aggregate influence propagated through a_v by $\delta \cdot \mathcal{T}_{a_j \to a_v}$. Summing over at most n neighbors gives:

$$\text{BlastRadius}(a_v) \leq \sum_{a_j \in \mathcal{N}(a_v)} \delta \cdot \mathcal{T}_{a_j \to a_v} \leq n \cdot \delta \cdot \max_{a_j} \mathcal{T}_{a_j \to a_v}. \tag{69}$$

This is a per-neighbor aggregate bound; it does not multiply the neighbor and reachable-agent counts. □

Corollary (Isolation Reduces Blast Radius)

If a_v has degree $|\mathcal{N}(a_v)| = k$, then:

$$\textit{BlastRadius}(a_v) \leq k \cdot \delta \cdot \max_{a_j} \mathcal{T}_{a_j \to a_v} \tag{70}$$

Restricting the degree of agents (least-privilege communication topology) proportionally reduces blast radius.

Updated Summary of Proof Techniques

Updated Summary of Proof Techniques I

Table 3: Summary of proof techniques by theorem (v1.1 additions marked with †).

Theorem	Primary Technique	Complexity	Edition
Trust Boundedness	Strong induction	$O(d)$	v1
Belief Injection Resistance	Probability independence	$O(1)$	v1
No Trust Amplification	Strong induction	$O(k)$	v1
Goal Alignment Invariant	Induction on time	$O(t)$	v1
Firewall Liveness	Complement probability	$O(1)$	v1
Byzantine Consensus	Classical BFT	$O(f)$	v1
Defense Composition Semiring†	Boolean algebra, lattice theory	$O(1)$	v2
Composition Detection Rate†	Product formula induction	$O(k)$	v2
FR Stealth-Impact Tight Bound†	Differential geometry (Fisher-Rao)	$O(n)$	v2
Agent Blast Radius Bound†	Graph traversal + trust bound	$O(n)$	v2
CIF-AD Full Coverage†	Matrix column inspection	$O(1)$	v2
OODA Security Invariant†	Product formula + induction	$O(k)$	v2
Adversary Class Separation†	Information theory (KL > 0)	$O(1)$	v2

Updated Summary of Proof Techniques I

All proofs in this supplement are constructive and provide explicit constants suitable for system implementation, parameter selection, and security analysis.